Once the command is executed successfully, the `cars` index is available instantly. Similar to the `_close` API, the `_open` API can be invoked on multiple indexes, including specifying indexes using wildcards.

So far, we've worked with several index operations and created indexes with custom mappings and settings, albeit individually. While this method is effective in development, it is far from ideal in production environments—it can be cumbersome to create a set of indexes over and over. Also, we may not want engineers to create indexes with an unexpected number of shards or replicas, which could destabilize our cluster. This is where we can use index templates.

Tighter control and business standards are expected as part of an overall indexing strategy, so Elasticsearch provides index templates to assist in developing indexes with an organizational strategy in mind. We can use this feature to apply configurations at scale, as discussed in the next section.

6.6 Index templates

Copying the same settings across various indexes, especially one by one, is tedious and sometimes error-prone. If we define an indexing template with a schema up front, a new index will be implicitly molded from this schema if the index name matches the template. Thus, any new index will follow the same settings and be homogenous across the organization, and DevOps won't need to repeatedly advocate for optimal settings with individual teams in the organization.

One use case for an indexing template is creating a set of patterns based on environments. For example, indexes for a development environment may have 3 shards and 2 replicas, while indexes for a production environment have 10 shards with 5 replicas each.

With index templating, we can create a template with predefined patterns and configurations. We can have a set of mappings, settings, and aliases bundled up in this template, along with an index name. Then, when we create a new index, if the index name matches the pattern name, the template is applied. Additionally, we can create a template based on a *glob* (global command) pattern, such as wildcards, prefixes, and so on.

> **NOTE** The *glob* pattern is commonly used in computer software to represent filename extensions (*.txt, *.cmd, *.bat, and so on).

Elasticsearch upgraded its template functionality beginning with version 7.8, and this newer version is abstract and much more reusable. (If you are interested in index templates from prior versions, see the official documentation for details.) Index templates are classified into two categories: *composable* index templates (or simply *index templates*) and *component* templates. As the name indicates, composable index templates are composed of zero or more component templates. An index template can also exist on its own without being associated with any component templates. Such index templates can contain all the required template features (mappings, settings, and aliases). They are used as independent templates when creating an index with a pattern.